Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Prognostic Data- and AI-based Mobility Impact Analyzer*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (x), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Greenhouse Gas Processing	*Non-CO2 Processing Technologies*	*Green transportation systems*	*IT convergence*
Concept of Proposed Technology	Aim of the proposed project is to develop a dashboard that evaluates emission savings from mobility services, such as fleet optimization and on-demand shuttles, by analyzing changes in user behavior. Key features include a user-friendly interface, real-time data integration, and scenario analysis tools. The dashboard utilizes data sources like service utilization statistics and emission factors to establish a baseline and forecast potential savings. Targeted at corporates and sustainability managers, it seeks to enhance decision-making and promote awareness of sustainable mobility solutions. Ultimately, it aims to provide insights into the effectiveness of mobility services in reducing emissions. Businesses should consider utilizing this dashboard tool as it empowers them to make informed, data-driven decisions regarding their mobility services. The integration of emission modeling is crucial, as it provides businesses with a clear understanding of their environmental impact and the effectiveness of various mobility strategies. By analyzing potential emission savings through accurate modeling, companies can identify opportunities for optimization that not only reduce costs but also enhance their sustainability profiles. This aspect is increasingly relevant as consumers and regulators demand greater accountability in environmental practices. Furthermore, the dashboard creates a significant market opportunity for entrepreneurs to develop software solutions and data-driven services tailored to mobility and transport management. These services can include advanced analytics platforms that utilize emission models, predictive modeling tools, and customized reporting solutions, all aimed at helping organizations effectively manage their emissions. As the demand for sustainable practices continues to grow, this dashboard could position businesses at the forefront of an emerging market focused on innovative mobility solutions, ensuring both compliance with regulations and alignment with corporate sustainability goals. The proposed project builds on the insights and algorithms developed in a previous project on the example of over 20 different mobility projects within a funding program of the Germany Ministry of Transportation that focused on			

(max 10 pages)

	impact simulation and emission calculations of model mobility measures to strengthen public transport (ÖPNV). By leveraging these advanced approaches, the aim is to achieve more accurate predictions of emission savings and changes in mobility behavior. The knowledge gained from this earlier work provides a solid foundation for developing innovative solutions for sustainable mobility and further enhancing the efficiency of public transportation systems.
Rationale for Joint R&D Project	- Global Challenges: The project targets climate change by evaluating emission savings and other sustainability indicators, which requires international collaboration to develop effective and credible solutions that adapt to different cultural backgrounds. - Diverse Perspectives: Engaging global partners introduces varied experiences, cultural backgrounds and innovative ideas, which can enhance the dashboard's design, functionality, and scalability to different markets. - Resource Sharing: Collaboration facilitates access to diverse data sources and technological expertise, improving the dashboard's real-time data integration and analysis capabilities. - Credibility through Scientific Rigor: A scientific approach ensures that the dashboard's analyses and predictions are based on robust methodologies, enhancing the credibility and reliability of the findings for stakeholders. - Long-Term Insights: Systematic examination of user behavior, grounded in scientific methods, provides valuable insights, ensuring that the project contributes to effective and sustainable mobility practices. - Proactive Response to Global Environmental Regulations: To comply with strengthening international carbon regulations, such as the EU Carbon Border Adjustment Mechanism (CBAM), collaboration with global research institutes is vital for securing objective carbon estimation models and technical validation.
Key Words	- Emission Reduction Modeling - User Behavior Analytics - Impact Simulation - Sustainability Data Metrics - Adaptive Demand Management

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4/5)	**End of the Project**	Level (8/9)
R&D Objectives	- Predictive Analytics Engine: An advanced algorithmic framework that forecasts emission reductions based on changes in user mobility behavior. - Real-Time Data Integration Framework: A robust system that connects and aggregates live data from various mobility services, enabling up-to-date analysis. - Dynamic Visualization Dashboard: An interactive platform that provides graphical representations of data trends, allowing users to explore metrics and insights intuitively. - Behavioral Modeling Tools: Software ready models designed to simulate and assess user mobility patterns, helping to identify potential impacts of different mobility strategies. - Cloud-Based Architecture: A scalable and flexible infrastructure that supports data hosting and application performance, ensuring accessibility for users across multiple devices. - Integrated Control Interface: Develop a cloud-based management tool for SMEs and public agencies to monitor and control carbon reduction status and fleet operations.			

(max 10 pages)

R&D Target Specifications	- Prediction Accuracy: Achieve at least 85% accuracy in forecasting emission reductions based on user behavior changes. - Data Integration Latency: Ensure real-time data integration with a latency of no more than 2 seconds for data updates from mobility services. - Data Processing Throughput: Process and analyze up to 1 million data points per minute, ensuring timely insights and reporting. - Carbon Emission Reduction: ≥12% reduction compared to non-optimized mobilty services
R&D Plan	Pre-Phase: Concept Development (6 Months) - Market Research and Needs Analysis: Identify the needs and requirements of potential users. - Definition of Project Goals: Establish clear objectives and requirements for the dashboard. - Development of Initial Prototypes: Create mockups to visualize the user interface. - Concept Validation: Gather feedback from stakeholders to refine the concept. - Creation of a Detailed Project Plan: Plan the next phases and allocate resources. Phase 1: Development (12 Months) - Implementation of the User Interface: Develop a user-friendly and intuitive interface. - Integration of Real-Time Data: Connect relevant data sources to support analyses. - Development of Emission Models: Create models for calculating and forecasting emission savings. - Conducting Tests: Ensure functionality and quality of the developed components. Phase 2: Pilot Testing (6 Months) - Conducting Pilot Projects: Test the dashboard with a selected group of users. - Feedback Collection: Analyze user experiences to identify areas for improvement. - Adjustments Based on Feedback: Implement changes to optimize the dashboard. Phase 3: Evaluation and Support (6 Months) - Evaluation of User Acceptance: Assess how well the dashboard is received by users. - Continuous Support: Offer technical support and maintenance services. - Planning for Updates: Develop strategies for future improvements and expansions. Phase 4: Market Launch (6 Months) - Finalization of the Dashboard: Complete development and prepare for the market entry. - Development of Marketing Strategies: Plan measures to promote the product. - User Training: Provide training materials and workshops for stakeholders. - Official Market Launch: Release the dashboard and initiate marketing efforts

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Algorithm Development: Design predictive algorithms for simulating the impacts of mobility measures on emissions. - Data acquisition for modelling: development and deploy of mobility surveys to collect the necessary data in the target markets for user behavior modelling. - Data Modeling and Analysis: Create models to represent user behavior and mobility patterns, and analyze data for insights on emission reductions. - Simulation of Mobility Measures: Conduct R&D simulations to evaluate the effectiveness of various mobility interventions on emissions reduction. - Prototype Development: Develop and test prototypes for the predictive analytics engine and visualization dashboard. - Validation and Testing: Perform validation of models and algorithms to ensure accuracy and reliability of predictions.

(max 10 pages)

Expected R&R of Korean SME Partner	- Data Provisioning: Supply historical and real-time data from mobility services necessary for analysis and modeling. - Stakeholder Engagement: Facilitate connections with end-users and other stakeholders to gather insights and validate findings. - Implementation of Solutions: Support the integration of developed technologies into existing operations and mobility services. - Feedback and Iteration: Provide ongoing feedback during the development process to refine algorithms and dashboard features. - Promotion and Training: Assist in promoting the final dashboard and conduct training sessions for users to ensure effective utilization. - Commercialization Strategy: Integrate the developed technologies into existing products and services to enhance their offerings and create new revenue streams. -

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Achievements: - Innovative Predictive Algorithms: Development of advanced algorithms that accurately forecast emission reductions based on mobility behavior changes and cultural background, enhancing decision-making capabilities. - Dynamic Visualization Dashboard: Creation of an intuitive dashboard that provides real-time insights into mobility patterns and emissions, facilitating user engagement and data-driven strategies. Commercialization Potential - Integration into Existing Products: The developed technologies can be seamlessly incorporated into the company's current offerings, enhancing their value proposition and attracting new customers. - New Revenue Streams: The dashboard and predictive analytics tools could be marketed as standalone products or services, creating additional income opportunities for the company. Broader Industrial Impact - Sustainability Advancements: The project will contribute to more sustainable transportation solutions, reducing emissions and promoting eco-friendly practices within the industry. - Enhanced Transport Efficiency: By optimizing mobility services, the project can improve transportation efficiency, leading to increased ridership and reduced congestion. Economic Impact - Cost Savings for Companies: Improved mobility strategies can lead to significant cost savings for businesses through optimized fleet operations and reduced emissions-related expenses. - Job Creation: The development and commercialization of these technologies may stimulate job growth in tech and transportation sectors, contributing to economic development. Social Impact: - Contribution to the early achievement of carbon neutrality goals by reducing urban traffic congestion, fine dust, and greenhouse gas emissions, thereby improving the quality of urban life. -

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6. References

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